Language of logic

1 Write down the symbol that corresponds to the following:

a Negation **b** Conjunction **c** Disjunction **d** Conditional

e Biconditional

2 Let

- p: A zebra has a mane.
- q: A dog can bark.

Write the following statements in symbolic form:

a A zebra does not have a mane.

b A zebra has a mane and a dog can bark.

c A dog cannot bark or a zebra does not have a mane.

d A dog cannot bark if, and only if, a zebra does not have a mane.

e If a zebra does not have a mane, then a dog cannot bark.

f A dog cannot bark, but a zebra has a mane.

3 Let

- p: The lemonade is sour.
- q: The rice is hot.

Write the following statements in symbolic form:

a The rice is not hot if, and only if, the lemonade is sour.

b If the lemonade is sour, then the rice is not hot.

c The lemonade is not sour, but the rice is hot.

d Neither is the lemonade sour nor is the rice hot.

e It is false that the lemonade is sour or the rice is hot.

f It is false that if the rice is not hot, then the lemonade is sour.

4 Let

- p: Dave has a laptop.
- q: James has a smartphone.

Write the following statements in words:

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a $\sim q$ **b** $p \wedge q$ **c** $\sim p \implies q$ **d** $\sim p \vee \sim q$

e $\sim (p \wedge q)$

5 Let

- p: Quentin has a rabbit.
- q: Maximilian has a dog.

Write the following statements in words:

a $\sim p$ **b** $p \vee q$ **c** $\sim p \iff \sim q$ **d** $\sim (p \vee q)$

e $\sim p \wedge \sim q$

6 Let

- p: The temperature is 90°.
- q: The heater is working.
- r: The apartment is cold.

Write the following statements in symbolic form:

a The temperature is 90° and the heater is not working, and the apartment is cold.

b The temperature is not 90° and the heater is working, but the apartment is cold.

c The temperature is 90° and the heater is working, or the apartment is cold.

d If the temperature is 90°, then the heater is working or the apartment is not cold.

7 Let

- p: The temperature is 80°.
- q: The fan is working.
- r: The apartment is hot.

Write the following statements in symbolic form:

a If the apartment is hot and the fan is working, then the temperature is 80°.

b The temperature is not 80° if, and only if, the fan is not working, or the apartment is not hot.

c The apartment is hot if, and only if, the fan is working, and the temperature is 80°.

d If the fan is working, then the temperature is 80°, if and only if, the apartment is hot.

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8 Let

- p: The water is $82^\circ\mathrm{F}$.
- q: The sun is down.
- r: We go swimming.

Write the following statements in words:

a $(p \vee q) \wedge \sim r$ **b** $\sim p \wedge (q \vee r)$ **c** $\sim r \implies (q \wedge p)$ **d** $(q \implies r) \wedge p$

e $(q \iff p) \wedge r$

9 Let

- p: The water is $60^\circ\mathrm{F}$.
- q: The sun is up.
- r: We go snorkelling.

Write the following statements in words:

a $(p \wedge q) \vee r$ **b** $(q \implies p) \vee r$

c $(q \wedge r) \implies p$ **d** $\sim p \implies (q \vee r)$

e $q \implies (p \iff r)$

10 Consider the statement: "Vincent started the fire and Valerie did not forget the meat".

a Write the statement in symbolic form by letting:

- p: Vincent started the fire.
- q: Valerie forgot the meat.

b What type of statement is this?

11 Consider the statement: "If the meeting is in Sydney, then we can see a film or we can go to the concert".

a Write the statement in symbolic form by letting:

- p: The meeting is in Sydney.
- q: We can see a film.
- r: We can go to the concert.

b What type of statement is this?

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12 Consider the statement: "It is false that if you eat healthy foods, then you will not gain fat".

a Write the statement in symbolic form by letting:

- p: You eat healthy foods.
- q: You gain fat.

b What type of statement is this?

13 Consider the statement: "If lunch is ready then we can eat, or we cannot go to the restaurant".

a Write the statement in symbolic form by letting:

- p: Lunch is ready.
- q: We can eat.
- r: We can go to the restaurant.

b What type of statement is this?

14 Consider the statement: "If the food has salt or the food has fat, then you will like it".

a Write the statement in symbolic form by letting:

- p: The food has salt.
- q: The food has fat.
- r: You will like the food.

b What type of statement is this?

15 Consider the statement: "If Yvonne is the instructor then Joanne is in Science class, if and only if, it is not a public holiday."

a Write the statement in symbolic form by letting:

- p: Yvonne is the instructor.
- q: Joanne is in Science class.
- r: It is a public holiday.

b What type of statement is this?

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16 Consider the statement: "If the motorbike has fuel and the battery has power, then the motorbike will start".

a Write the statement in symbolic form by letting:

- p: The motorbike has fuel.
- q: The battery has power.
- r: The motorbike will start.

b What type of statement is this?

17 Consider the statement: "The movie theater is closed if, and only if, it is a public holiday, or it is before 7 am.

a Write the statement in symbolic form by letting:

- p: The movie theater is closed.
- q: It is a public holiday.
- r: It is before 7 am.

b What type of statement is this?

Additional questions

18 What is a biconditional statement and how can it be recognized?

19 Which of the following statements are biconditional?

a A triangle is equilateral if and only if it is equiangular.

b If I am at South Beach, then I am next to the ocean.

c It is Cinco de Mayo if and only if it is the 5th of May.

d Tangents from a point to the same circle are congruent.

20 Consider the statement:
"A ray is an angle bisector if and only if it divides an angle into two congruent angles."
State the two conditional statements that make up this biconditional statement.

21 Consider the two theorems:
"If two sides of a triangle are congruent, then the angles opposite them are congruent."
"If two angles of a triangle are congruent, then the sides opposite them are congruent."
Write a biconditional statement combining these two theorems.

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22 What are the requirements for a biconditional statement to be true?

23 For each of the following biconditional statements, explain whether they are true or false. If false, justify your answer with a counterexample.

- **a** A number is even if and only if it is divisible by two.
- **b** A real number is greater than three if and only if it is greater than four.
- **c** A number is prime if and only if it has no factors other than itself and one.
- **d** A shape is a regular polygon if and only if all its sides are congruent.

24 Suppose the following statement is true.
"A quadrilateral is a parallelogram if and only if its diagonals bisect each other."
Write the two conditional statements that must be true.

25 Write each of the following definitions as a biconditional statement.

- **a** A triangle median is a segment whose endpoints are a vertex and the midpoint of the opposite side.
- **b** A right angle is an angle measuring exactly 90°.
- **c** A sphere is a three-dimensional figure consisting of all points equidistant from a point called the center.

26 Explain why each biconditional statement is not true.

- **a** Neva will use an umbrella if and only if it raining.
- **b** The moon is visible if and only if we cannot see the sun.
- **c** Deloris walks to school if and only if she misses the bus.
- **d** Ulrich can get items from the top shelf of a cupboard if and only if he is tall.

27 Determine whether each of the following statements is a valid definition or not. In not, explain why.

- **a** If an angle is obtuse, then its measure is greater than 90°.
- **b** Congruent segments are segments with the same length.
- **c** Adjacent angles are next to each other.
- **d** Perpendicular lines have a vector dot product equal to zero.

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28 Suppose that a biconditional statement is not true.
State if we can determine which of the implied conditional statements are true or false.
Explain your conclusion.

29 Explain why a valid definition can be written as a biconditional statement.

30 Which value for x is a counterexample to the following statement?
For all positive values of x, $x^3 \geq x$

- $^{-}1.0$
- $^{-}0.1$
- 0.1
- 1.0

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